

rumors dealing with space sustainability, debris, law, resource utilisation, space-based navigation, telecommunications, trade, and commerce.

Emerging technologies play a vital role in reducing the cost of access to space and improving mission efficiency. For instance, our upcoming NGLV (Next-Generation Launch Vehicle) will feature partial reusability and advanced manufacturing techniques, including additive and inline manufacturing, to lower production costs and shorten turnaround times.

In quantum technologies, Isro is working under the National Quantum Mission to develop quantum key distribution (QKD) satellites for secure communication. AI is already being utilised for tasks such as image analysis, object classification, and mission design. In the future, it will have a broader role in satellite operations, system design, and data processing, with onboard AI enhancing satellite autonomy.

Isro actively contributes to global efforts by sharing critical climate and weather data from our satellites. In disaster management, Isro supports global rescue operations and early warning systems.

We provide data across many parameters, including sea states, sea surface temperatures, ice, glaciers, and ice sheets, among others. In the coming days, we are focussing on the synthetic aperture radar. This satellite, designed with a focus on climate, weather, and sustainability, will map the entire globe every 12 days. Under the TRISHNA mission, we will provide thermal imaging of the Earth.

We don't differentiate between Isro and private entities; we view this as a national space ecosystem where Isro plays an enabling role. There is no competition between the two; instead, it's about fostering organic growth. Isro's goal is to develop a capable private sector that can handle end-to-end tasks. Currently, private players largely act as subcontractors,

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but we aim to transition operational tasks to them as they mature. To support start-ups, we have established a ₹1,000-crore venture capital fund.

Technical parks are also being set up across the country to provide infrastructure for testing and evaluation. Additionally, Isro's facilities and knowledge are now open to private companies for mentoring and support. This ensures that the knowledge and resources within Isro contribute to the growth of the private sector.

My vision revolves around scaling up the space economy. While technological achievements like landing on the Moon are crucial, we also focus on making space activities economically significant. Historically, this sector has relied heavily on government funding. By scaling up and involving private players, we aim to transition to a model where the economic impact of space activities becomes self-sustaining.

However, this is challenging. Space ventures are capital-intensive, have long gestation periods, and offer

relatively low returns compared to traditional industries.

Government sponsorship will remain essential for strategic programmes, but we envision a shift where private companies handle activities like communication networks, remote sensing applications, and infrastructure development. We also foresee significant downstream revenue generation—up to 70% of the space economy—coming from services like data applications, product generation, and communication systems. The government will focus on R&D and ambitious missions, such as human spaceflight and lunar exploration, while operational responsibilities increasingly transition to private players.

The Indian space industry has evolved significantly. Private companies, once hesitant to engage in space ventures without Isro's support, are now venturing into new domains independently. Our vision is for Indian companies to leverage global technologies, create world-class products, and establish themselves as global players. The transformation is underway, and the collaboration between Isro, private enterprises, and start-ups will drive India's space ambitions to the next frontier. **BT**

As told to Neetu Chandra Sharma. Views are personal

| 75

